

Math 300 – Fake Midterm

Rutgers University

June 27, 2023

- Please answer all questions within the designated space. Additional pages are provided at the end of the exam for any supplementary work.
- Neatness and organization matter. Please request extra paper for draft work if needed it.
- This exam is closed-book, closed-notes.
- You are not allowed to use calculators or any electronic devices on the exam.
- This exam consists of 8 questions for a total of 150 points, and a bonus question worth 15 points.

Name: _____

NetID: _____

Question	Points	Bonus Points	Score
1	15	0	
2	10	0	
3	10	0	
4	15	0	
5	25	0	
6	25	0	
7	25	0	
8	25	0	
9	0	15	
Total:	150	15	

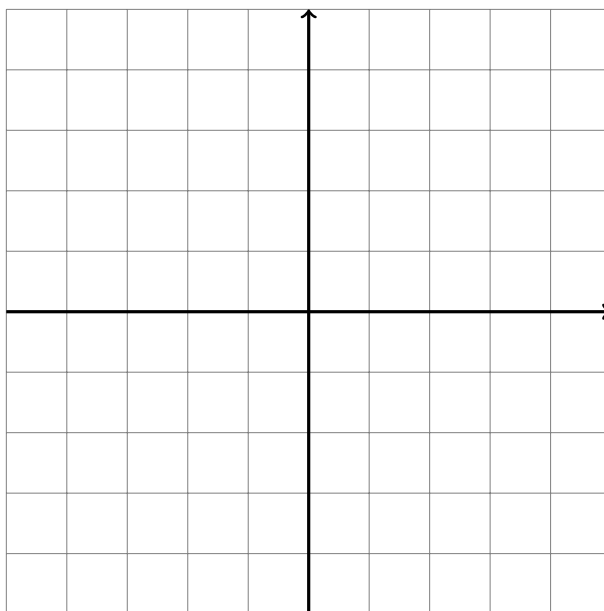
1. Given the indexed sets $A_n = \{k \in \mathbb{Z} : k \geq n\}$ and $B_n = \{k \in \mathbb{Z} : k \leq n\}$, write the following sets by listing their elements between braces.

(a) (5 points) $A_0 \cup B_0$

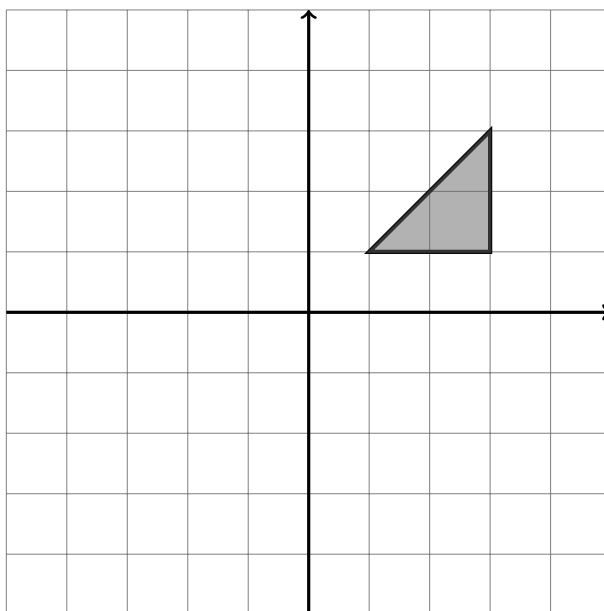
(b) (5 points) $A_{2023} \cap B_{2025}$

(c) (5 points) $(\overline{A_{2025}} \cap \overline{B_{2023}}) \times (A_0 \cap B_0)$

2. (a) (5 points) Sketch $X \cap Y$ for $X = \{(x, y) \in \mathbb{R}^2 : |y| \leq |x|\}$ and $Y = \{(x, y) \in \mathbb{R}^2 : x^2 \leq 4\}$.



- (b) (5 points) Given the sketch below, write the shaded set Z in set-builder notation.



3. (10 points) Use a truth table to verify if the following statements are logically equivalent. Make sure to state your conclusion.

$$“(P \implies Q) \vee R” \text{ and } “\sim ((P \wedge \sim Q) \wedge \sim R)”$$

4. Negate all the following statements. Words to words, symbols to symbols.

(a) (5 points) Every differentiable function is continuous.

(b) (5 points) $\forall \varepsilon > 0, \exists \delta > 0, (|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon).$

(c) (5 points) The equation $x^n + y^n = z^n$ has an integer solution for some $n > 3$.

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5. (25 points) Prove the true statement and provide a counterexample for the false statement. Suppose that x and y are non-negative real numbers.
- (a) The geometric mean is always less or equal than the arithmetic mean.
 - (b) The geometric mean is always less or equal than the harmonic mean.

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6. (25 points) Suppose a is an integer. If a is odd, then $x^2 + x - a^2 = 0$ has no integer solution.

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7. (25 points) Prove that for all $x \in \mathbb{R}$, $\sqrt{2} - x$ and $\sqrt{2} + x$ cannot both be rational numbers.

8. (25 points) Let $S = \{1, 2, 3, \dots, 2n\}$. Show that if we choose $n + 1$ numbers from S , then there exist two numbers such that one is a multiple of the other.

9. (15 points (bonus)) A *lattice point* in $\mathbb{R}^2$ is a point (x, y) whose coordinates x and y are integers. Suppose that every lattice point is colored either red, yellow or blue. Prove that there exists a rectangle in the plane such that all of its vertices are the same color.

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